LIFE SCIENCES

LIFE SCIENCES GRADE 12 TERM 3 CONTROLLED ASSESSMENT PACK

SUBJECT	LIFE SCIENCES
GRADE	12
ASSESSMENT TASK	LIFE SCIENCES GRADE 12 TERM 3 CONTROLLED ASSESSMENT PACK
MARKS	50
TIME	1 hour

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.

2. Answer all questions or complete all activities as instructed.

3. Show all working where calculations, planning, diagrams or explanations are required.

4. Use the space provided by the educator, answer booklet or learner task book.

5. Write neatly and clearly.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Data Table

2.2 The table below shows the levels of two hormones during the menstrual cycle of a healthy human female.

DAY OF MENSTRUAL CYCLE	OESTROGEN LEVEL (pg/mℓ)	PROGESTERONE LEVEL (ng/mℓ)
4	55	0,23
8	70	0,03
10	280	0,03
12	300	0,03
14	140	3,0
16	110	12,5
20	80	15,0
24	70	5,0
28	65	0,8

2.2.1 On which day of this menstrual cycle is the level of progesterone the highest? (1)

2.2.2 Name the reproductive hormone that will begin to increase from day 24 of this female's menstrual cycle. (1)

2.2.3 Use data from the table to explain your answer to QUESTION 2.2.2. (2)

2.2.4 Calculate the percentage increase in the oestrogen level from day 8 to day 10. Show ALL your working. (3)

2.2.5 How would the progesterone level differ after day 20 if this female was pregnant? (1)

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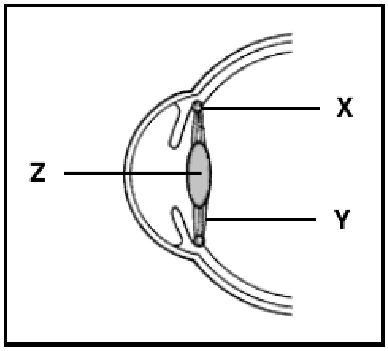
SOURCE 2: Official-paper exemplar: Diagram Or Model

1.1.6 The diagram below represents a part of the human eye.



A spectator in a soccer stadium is seated 200 metres away from the field.

Which ONE of the following describes the condition of structures **X**, **Y** and **Z** when he looks at the ball placed in the middle of the field?

	MUSCLE IN X	STRUCTURE Y	PART Z
A	Relaxed	Slack	More convex
B	Contracted	Slack	Less convex
C	Relaxed	Taut	Less convex
D	Contracted	Taut	More convex

1.1.7 Which ONE of the following is involved in thermoregulation?

A Corpus callosum
B Hypothalamus
C Cerebellum
D Spinal cord

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SOURCE 3: Official-paper exemplar: Graph Or Chart

2.3.4 Draw a bar graph to show the results in the table. (6)
(14)

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QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND DATA

Instructions: Use the supplied data, diagram or graph.

Stimulus: Data/diagram stimulus linked to Evolution by natural selection and speciation.

1.1. Define one key concept from the term focus. **(2)**

1.2. Describe the trend or relationship shown by the supplied data. **(5)**

1.3. Explain the process or principle using correct terminology. **(8)**

1.4. Calculate or infer a result from the supplied values and show working where needed. **(5)**

QUESTION 2: SECTION B: INVESTIGATION DESIGN

Instructions: Critique method, variables and evidence.

Stimulus: Investigation focus: Scientific investigation design for natural selection.

2.1. State a suitable aim or hypothesis for the investigation. **(3)**

2.2. Identify independent, dependent and controlled variables. **(6)**

2.3. Evaluate validity and reliability and suggest one improvement. **(6)**

QUESTION 3: SECTION C: APPLICATION

Instructions: Use subject knowledge and supplied evidence.

Stimulus: Application focus: Using data and anatomical evidence to support evolutionary explanations.

3.1. Answer an integrated explanation question using evidence from the stimulus. **(15)**

MARKING GUIDELINE

TASK 1: SECTION A: CONCEPTS AND DATA

1.1. Define one key concept from the term focus. **(2)**

- Definition scientifically accurate.

1.2. Describe the trend or relationship shown by the supplied data. **(5)**

- Trend described.
- Evidence cited.

1.3. Explain the process or principle using correct terminology. **(8)**

- Correct terms.
- Logical explanation.

1.4. Calculate or infer a result from the supplied values and show working where needed. **(5)**

- Working shown.
- Answer accurate.

TASK 2: SECTION B: INVESTIGATION DESIGN

2.1. State a suitable aim or hypothesis for the investigation. **(3)**

- Aim/hypothesis testable.

2.2. Identify independent, dependent and controlled variables. **(6)**

- Variables correctly classified.

2.3. Evaluate validity and reliability and suggest one improvement. **(6)**

- Validity/reliability explained.
- Improvement practical.

TASK 3: SECTION C: APPLICATION

3.1. Answer an integrated explanation question using evidence from the stimulus. **(15)**

- Evidence used.
- Explanation coherent.
- Conclusion supported.

RUBRIC

Criterion	Marks	Descriptor
Subject accuracy and terminology	12	Uses accurate concepts and terminology.
Evidence, data or source	12	Uses supplied evidence

use		appropriately.
Reasoning and explanation	12	Shows logical, grade-appropriate reasoning.
Communication and completion	14	Answers are clear, complete and structured.

EDUCATOR REVIEW CHECKLIST

- Confirm CAPS term and topic fit before classroom use.
- Insert or approve final source material where the pack requests educator-supplied sources.
- Check mark allocation, memo wording and learner level.
- Confirm visuals are suitable and not treated as official source material unless sourced separately.